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Mathematical Programming

An Introduction to Optimization

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About the Course

- Credit hours: 48
- Textbook: *Mathematical Programming: An Introduction to Optimization*, by Melvyn W. Jeter
- Reference: *Convex Optimization*, by Stephen Boyd
- Performance Assessment:
 - 1 Class attendance (10%)
 - 2 Assignments (40%)
 - 3 Final Exam (50%)



Acknowledgments

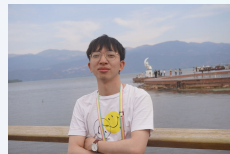
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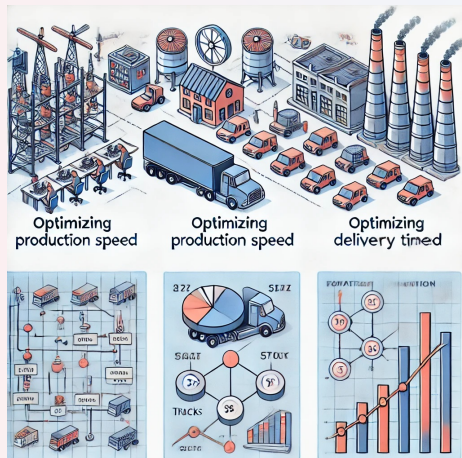


Introduction

- An introduction to optimization for students with background of linear algebra, calculus, and multivariable calculus
- Optimization is used everywhere nowadays with growing importance
- Applications:
 - Economic analysis
 - Management
 - Engineering technology
 - Military operations
 - etc



Examples



Examples



The Basic Mathematical Programming Problem

■ Definition:

$$\min f(x_1, x_2, \dots, x_n) \quad \text{or} \quad \max f(x_1, x_2, \dots, x_n) \quad (1)$$

$$\text{s.t.} \quad (x_1, x_2, \dots, x_n) \in \Omega \quad (2)$$

where

- $f(x_1, x_2, \dots, x_n)$ in (1) is the cost (objective) function.
- Ω in (2) is the set of feasible solutions.
- Unconstrained problem: $\Omega = \{(x_1, \dots, x_n) : x_i \in \mathbb{R}\}$.
- Constrained problem: Ω is a proper subset of $\{(x_1, \dots, x_n) : x_i \in \mathbb{R}\}$.



Linear Programming Problem

- Linear cost function:

$$f(x_1, \dots, x_n) = c_1x_1 + \dots + c_nx_n, \quad (3)$$

where $c_1, \dots, c_n$ are real constants. c_i is called the cost coefficient of the variable x_i .

- An equality constraint:

$$a_1x_1 + \dots + a_nx_n = b, \quad (4)$$

where $a_1, \dots, a_n$ are real constants.

- A linear inequality constraint:

$$a_1x_1 + \dots + a_nx_n \leq b, \quad (5)$$

- A problem that is NOT linear programming: nonlinear programming problem.
- Generally, linear programming problems are easier than nonlinear ones.



Example 1

- A manufacturer produces n products $P_1, P_2, \dots, P_n$ from m raw materials $M_1, \dots, M_m$. The amount of each raw material required for each product:

	P_1	P_2	$\dots$	P_n
M_1	$a_{1,1}$	$a_{1,2}$	$\dots$	$a_{1,n}$
M_2	$a_{2,1}$	$a_{2,2}$	$\dots$	$a_{2,n}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
M_m	$a_{m,1}$	$a_{m,2}$	$\dots$	$a_{m,n}$

- Available: b_i units of M_i , $\forall i = 1, \dots, m$.
- Revenue: c_j dollars for each unit of P_j sold.
- Question: Desire to maximize the total daily revenue.



Example 1

- x_j : # of product P_j that has been produced.
- $a_{i,j}x_j$: # of material M_i used to produce x_j units of P_j .
- # of the daily revenue:

$$c_1x_1 + \dots + c_nx_n \quad (6)$$

- # of material M_i consumed in total:

$$a_{i,1}x_1 + \dots + a_{i,n}x_n \leq b_i, \quad (7)$$

Example 1

- The cost function:

$$\max_{x_j, \forall j \in [1, n]} \{c_1 x_1 + \dots + c_n x_n\} \quad (8)$$

subject to

$$\begin{aligned} a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,n}x_n &\leq b_1, \\ a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,n}x_n &\leq b_2, \\ &\vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \dots + a_{m,n}x_n &\leq b_m, \\ x_1 \geq 0, x_2 \geq 0, \dots, x_n &\geq 0. \end{aligned} \quad (9)$$

Example 1

- Continuous variable: allowed to take on all fractional as well as integral values within its range of variation
- Discrete variable: must take on values from a discrete set of values
- Slack variable: inequality can be converted to an equality constraint by adding a non-negative variable
 - Take (9) for example:

$$a_{1,1}x_1 + \dots + a_{1,n}x_n + x_{n+1} = b_1,$$

x_{n+1} : Slack variable, assigned a cost coefficient of $c_{n+1} = 0$



Example 1

- The cost function (8) can be reformulated as

$$\max_{x_j, \forall j \in [1, n+m]} \underbrace{\{c_1 x_1 + \dots + c_n x_n + c_{n+1} x_{n+1} + \dots + c_{n+m} x_{n+m}\}}_{=\max_{x_j, \forall j \in [1, n]} \{c_1 x_1 + \dots + c_n x_n\}}$$

subject to

$$\begin{aligned} a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,n}x_n + x_{n+1} &= b_1, \\ a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,n}x_n + x_{n+2} &= b_2, \\ &\vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \dots + a_{m,n}x_n + x_{n+m} &= b_m, \\ x_1 \geq 0, \dots, x_n \geq 0, \quad x_{n+1} \geq 0, \dots, x_{n+m} &\geq 0. \end{aligned}$$

Example 2

- 4 salesmen + 4 sales districts
- The sales productivity ratings for each salesman S_i in each sales districts D_j :

	D_1	D_2	D_3	D_4
S_1	5	10	8	2
S_2	3	4	2	6
S_3	9	5	4	5
S_4	5	5	6	4

- Question: Determine the best assignment of salesman to sales districts.



Example 2

- Define

$$x_{i,j} = \begin{cases} 1, & \text{if } S_i \text{ assigned to } D_j, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

- The value of an assignment:

$$\begin{aligned} f(x_{i,j}) = & 5x_{1,1} + 10x_{1,2} + 8x_{1,3} + 2x_{1,4} + 3x_{2,1} + 4x_{2,2} + 2x_{2,3} + 6x_{2,4} \\ & + 9x_{3,1} + 5x_{3,2} + 4x_{3,3} + 5x_{3,4} + 5x_{4,1} + 5x_{4,2} + 6x_{4,3} + 4x_{4,4}. \end{aligned}$$

- Each salesman must be assigned to exactly one district:

$$\sum_{j=1}^4 x_{i,j} = 1, \quad \forall i. \quad (11)$$

- Each district is assigned exactly one salesman:

$$\sum_{i=1}^4 x_{i,j} = 1, \quad \forall j. \quad (12)$$

Example 2

■ Problem:

$$\max_{\forall i,j} \{f(x_{i,j})\}$$

subject to

$$\begin{aligned}x_{i,1} + x_{i,2} + x_{i,3} + x_{i,4} &= 1, & \forall i, \\x_{1,j} + x_{2,j} + x_{3,j} + x_{4,j} &= 1, & \forall j, \\0 \leq x_{i,j} &\leq 1, & \forall i, j.\end{aligned}$$

Example 3

- A flour company has 3 production lines which can produce A and B. The number of ten pound sacks of flour which can be produced daily by each of the production lines is summarized in the following table:

	L_1	L_2	L_3
A	15000	10000	12000
B	12000	13500	14500

- Question: Produce as many ten pound bags of flour of brands A and B as possible.



Example 3

- Define

$$x_i = \begin{cases} 1, & \text{if } L_i \text{ assigned to A,} \\ 0, & \text{otherwise,} \end{cases} \quad w_i = \begin{cases} 1, & \text{if } L_i \text{ assigned to B,} \\ 0, & \text{otherwise.} \end{cases}$$

- A production line cannot be assigned to both products:

$$x_1 w_1 = 0, \quad x_2 w_2 = 0, \quad x_3 w_3 = 0.$$

- All variables are non-negative



Example 3

- The above two constraints can be compressed into one:

$$\sum_{i=1}^3 x_i w_i = x_1 w_1 + x_2 w_2 + x_3 w_3 = 0. \quad (13)$$

- To insure that each production line is assigned to a job:

$$x_1 + w_1 = 1,$$

$$x_2 + w_2 = 1,$$

$$x_3 + w_3 = 1.$$

Example 3

- A nonlinear programming problem:

$$\max_{\forall x_i, w_j} \{15000x_1 + 10000x_2 + 12000x_3 + 12000w_1 + 13500w_2 + 14500w_3\}$$

subject to

$$x_1 + w_1 = 1,$$

$$x_2 + w_2 = 1,$$

$$x_3 + w_3 = 1,$$

$$\sum_{i=1}^3 x_i w_i = 0, \quad (\text{complementary constraint})$$

$$0 \leq x_i \leq 1, \quad 0 \leq w_i \leq 1, \quad \forall i.$$

Example 4

- Quadratic (nonlinear) programming problem:

$$\max_{\forall i,j} \left\{ c_1 x_1 + \dots + c_n x_n + \sum_{i=1}^n \sum_{j=1}^n b_{ij} x_i x_j \right\}$$

subject to

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1,$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2,$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m,$$

$$x_1 \geq 0, \quad x_2 \geq 0, \dots, \quad x_n \geq 0.$$

Example 4

- The portfolio selection problem:

$$\max_{\forall i,j} \left\{ c_1 x_1 + \dots + c_n x_n - \sum_{i=1}^n \sum_{j=1}^n b_{i,j} x_i x_j \right\}$$

subject to

$$x_1 + \dots + x_n = 1,$$

$$x_1 \geq 0, \dots, x_n \geq 0.$$

- x_i : investment
- c_i : expected rate of return
- $\sum_{i=1}^n \sum_{j=1}^n b_{i,j} x_i x_j$: risk of investment



Global And Local Solutions

- Standard Euclidean distance:

$$\|x - \bar{x}\| = \sqrt{\sum_{i=1}^n (x_i - \bar{x}_i)^2}$$

- Neighborhood of $\bar{x}$:

$$N_{\varepsilon}(\bar{x}) = \{x : \|x - \bar{x}\| < \varepsilon\}$$

where ε is a positive real number.



Global And Local Solutions

- Local minimum (at $\bar{x} \in \Omega$):

$$\exists \varepsilon > 0, \text{ such that } f(\bar{x}) \leq f(x), \quad \forall x \in \Omega \cap N_\varepsilon(\bar{x}).$$

- Strict local minimum:

$$\exists \varepsilon > 0, \text{ such that } f(\bar{x}) < f(x), \quad \forall x \in (\Omega \cap N_\varepsilon(\bar{x})) \setminus \{\bar{x}\}.$$

- Global minimum:

$$f(\bar{x}) \leq f(x), \quad \forall x \in \Omega.$$

- Strict global minimum:

$$f(\bar{x}) < f(x), \quad \forall x \in \Omega \setminus \{\bar{x}\}.$$



Topology Concepts

- The interior of Ω :

$$\text{Int}(\Omega) = \{x \in \Omega : N_\varepsilon(x) \subseteq \Omega \text{ for some } \varepsilon > 0\}.$$

- Ω is open if $\Omega = \text{Int}(\Omega)$, i.e., every point of Ω is an interior point

- The exterior of Ω :

$$\text{Ext}(\Omega) = \{x : N_\varepsilon(x) \subseteq E^n \setminus \Omega \text{ for some } \varepsilon > 0\}$$

where

$$E^n = \{x = (x_1, \dots, x_n) : x_i \in \mathbb{R}\},$$
$$E^n \setminus \Omega = \{x \in E^n : x \notin \Omega\}.$$



Topology Concepts

- x is a boundary point of Ω :

$$\exists x_1 \in \Omega : x_1 \in N_\varepsilon(x), \quad \forall \varepsilon > 0,$$

$$\exists x_2 \notin \Omega : x_2 \in N_\varepsilon(x), \quad \forall \varepsilon > 0.$$

- The boundary $\text{Bd}(\Omega)$:

The collection of all boundary points of Ω

- A set Ω is closed:

$$\text{Bd}(\Omega) \subseteq \Omega$$

- x is an accumulation point of Ω :

$$\exists x' \neq x, \quad x' \in \Omega : \quad x' \in N_\varepsilon(x), \quad \forall \varepsilon > 0.$$



Post-Optimal (Sensitivity) Analysis

- Data in a mathematical programming problem can change as conditions change: Data
→ Estimates/Measurements
- Possible to solve new problem by modifying the original one, thus avoiding completely solving the new problem
- Post-optimal analysis:

optimal solution of
existing problem



optimal solution of
modified problem



Example 5

- A company's products are stored in m places, and sold in n outlets
- Shipping cost from the i -th storage to the j -th outlet: $c_{i,j}$ dollars
- One day, each outlet requires exactly d_j units of product
- Each storage has s_i units of product available for shipment
- Question: To minimize the shipping cost, determine the number of units should be shipped from each storage to each outlet



Example 5

- Question: To minimize the shipping cost, determine the number of units should be shipped from each storage to each outlet

$$\min_{x_{i,j}} \sum_{i=1}^m \sum_{j=1}^n c_{i,j} x_{i,j}$$

subject to

$$\sum_{j=1}^n x_{i,j} \leq s_i, \quad \forall i,$$

$$\sum_{i=1}^m x_{i,j} = d_j, \quad \forall j,$$

$$x_{i,j} \geq 0, \quad \forall i, j.$$

Parametric Programming Problem

- Question: Product is good. Customers want more. Outlet requires an increase of $\bar{d}_j$ per day, and after t days

$$\min_{x_{i,j}} \sum_{i=1}^m \sum_{j=1}^n c_{i,j} x_{i,j}$$

subject to

$$\sum_{j=1}^n x_{i,j} \leq s_i, \quad \forall i,$$

$$\sum_{i=1}^m x_{i,j} = d_j + t\bar{d}_j, \quad \forall j,$$

$$x_{i,j} \geq 0, \quad \forall i, j,$$

$$t \geq 0.$$



Model Stability

- Model Stable: small changes in the data $\rightarrow$ small changes in $x_1^*, \dots, x_n^*$ and $f(x_1^*, \dots, x_n^*)$
- Model Unstable: small changes in the data $\rightarrow$ LARGE changes in $x_1^*, \dots, x_n^*$ and $f(x_1^*, \dots, x_n^*)$
- Frequently, model-unstable problems possess redundant constraints

$$g_i(x_1, \dots, x_n) = b_i, \quad \forall i,$$

$$\sum_{i=1}^m \lambda_i g_i(x_1, \dots, x_n) = \sum_{i=1}^m \lambda_i b_i$$

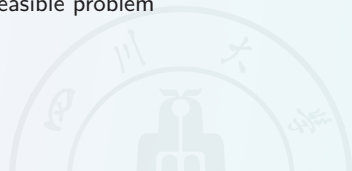
Feasible problem

$\Rightarrow$

$$g_i(x_1, \dots, x_n) = b_i, \quad \forall i,$$

$$\sum_{i=1}^m \lambda_i g_i(x_1, \dots, x_n) = \varepsilon + \sum_{i=1}^m \lambda_i b_i$$

Infeasible problem



A Review of Linear Algebra: Matrix

■ Matrix:

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{bmatrix}$$

■ Compact expression: $A = [a_{i,j}]$

- $a_{i,j}$: the (i,j) -th entry of A
- $\mathbb{R}^{m \times n}$: The collection of all $m \times n$ real matrices



A Review of Linear Algebra: Matrix

- The transpose of A :

$$A^T = \begin{bmatrix} a_{1,1} & a_{2,1} & \cdots & a_{m,1} \\ a_{1,2} & a_{2,2} & \cdots & a_{m,2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1,n} & a_{2,n} & \cdots & a_{m,n} \end{bmatrix}$$



A Review of Linear Algebra: Vector

- Column matrix (vector):

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} = [x_1, \dots, x_m]^T \in \mathbb{R}^{m \times 1}$$

- Row matrix:

$$y^T = [y_1, \dots, y_n] \in \mathbb{R}^{1 \times n}$$

- Scalars: real/complex numbers
- Vector space ($\mathbb{R}^{m \times 1}$): Euclidean m -space (E^m)



A Review of Linear Algebra: Vector

- Vector addition:

$$x + y = [x_1 + y_1, \dots, x_n + y_n]^T, \text{quad} x, y \in \mathbb{R}^{n \times 1}$$

- For a scalar α , scalar multiplication:

$$\alpha x = [\alpha x_1, \dots, \alpha x_n]^T$$



A Review of Linear Algebra: Vector

■ Inner product:

$$x^{\top}y = \langle x, y \rangle = \sum_{i=1}^n x_i y_i$$

■ Properties:

- $\langle x, y \rangle = \langle y, x \rangle$
- $\langle (\alpha x + \beta y), z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$
- $\langle x, x \rangle \geq 0$, with equality holds only when $x_i = 0, \forall i$
- Euclidean norm: $\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{x^{\top}x}$



A Review of Linear Algebra: Subspace

- Subspace S of $\mathbb{R}^{n \times 1}$: Any nonempty set of vectors, such that $x, y \in S$, $\alpha, \beta \in \mathbb{R}$, and $\alpha x + \beta y \in S$
- Linear combination:

$$\begin{aligned}\sum_{i=1}^k \alpha_i x_i &= \alpha_1 x_1 + \dots + \alpha_k x_k \\ &= \left[\sum_{i=1}^k \alpha_i x_{i,1}, \dots, \sum_{i=1}^k \alpha_i x_{i,n} \right]^T\end{aligned}$$

- If $B \subseteq \mathbb{R}^{n \times 1}$, all linear combination of elements of B :

$$L(B) \triangleq \left\{ \sum_{i=1}^k \alpha_i x_i \mid \alpha_i \in \mathbb{R}, x_i \in B, k \in \mathbb{N} \right\}$$



Example 5

Let $B_1 = \{[1, 0, 0]^T, [0, 1, 0]^T, [0, 0, 1]^T\}$.

$$\begin{aligned} L(B_1) &= \{x_1[1, 0, 0]^\top + x_2[0, 1, 0]^\top + x_3[0, 0, 1]^\top : x_1, x_2, x_3 \in \mathbb{R}\} \\ &= \{[x_1, x_2, x_3]^\top : x_1, x_2, x_3 \in \mathbb{R}\} \\ &= \mathbb{R}^{3 \times 1}. \end{aligned}$$

Example 6

Let $B_2 = \{[1, 0, 0]^T, [1, 0, 1]^T\}$.

$$\begin{aligned} L(B_2) &= \{x_1[1, 0, 0]^T + x_2[1, 0, 1]^T : x_1, x_2 \in \mathbb{R}\} \\ &= \{[x_1 + x_2, 0, x_2]^T : x_1, x_2 \in \mathbb{R}\}. \end{aligned}$$

$L(B_2)$ is the x_1 - x_3 plane.



Example 6

Let $B_3 = \{[1, 0, 0]^\top, [1, 0, 1]^\top, [4, 0, 1]^\top\}$.

$$\begin{aligned} L(B_3) &= \{x_1[1, 0, 0]^\top + x_2[1, 0, 1]^\top + x_3[4, 0, 1]^\top : x_1, x_2, x_3 \in \mathbb{R}\} \\ &= \{[x_1 + x_2 + 4x_3, 0, x_2 + x_3]^\top : x_1, x_2, x_3 \in \mathbb{R}\}. \end{aligned}$$

$L(B_3) = L(B_2)$, i.e., the x_1 - x_3 plane, again.



A Review of Linear Algebra: Subspace

- $L(B)$ is a subspace of $\mathbb{R}^{n \times 1}$: $L(B)$ is a subspace spanned (or generated) by B
- If $S = L(B)$, then B spans S
- ◇ Any intersection of subspaces of $\mathbb{R}^{n \times 1}$ is also a subspace of $\mathbb{R}^{n \times 1}$
- ◇ $L(B)$ is the intersection of all subspaces of $\mathbb{R}^{n \times 1}$ that contain $B \rightarrow L(B)$ is the “smallest” subspace that contains B

Example 7

Let $S = \{[x_1, x_2, x_3]^T : x_1 + x_2 = 0, x_1, x_2, x_3 \in \mathbb{R}\}$.

Question: Prove that S is a subspace of $\mathbb{R}^{3 \times 1}$, and find a set that spans S

Solution:

$$\begin{aligned} S &= \{[x_1, x_2, x_3]^T : x_1 + x_2 = 0, x_1, x_2, x_3 \in \mathbb{R}\} \\ &= \{[x_1, -x_1, x_3]^T : x_1, x_3 \in \mathbb{R}\} \\ &= \{x_1[1, -1, 0]^T + x_3[0, 0, 1]^T : x_1, x_3 \in \mathbb{R}\} \\ &= L(\{[1, -1, 0]^T, [0, 0, 1]^T\}). \end{aligned}$$

S is a subspace that is spanned by $\{[1, -1, 0]^T, [0, 0, 1]^T\}$.



A Review of Linear Algebra: Subspace

■ Definition:

$B \subseteq \mathbb{R}^{n \times 1}$, B is a linearly dependent set of vectors

$$\iff$$

$\exists$ distinct $x_1, \dots, x_k \in B$, $\alpha_1, \dots, \alpha_k \in \mathbb{R}$, not ALL $\alpha_i = 0$, such that

$$\alpha_1 x_1 + \dots + \alpha_k x_k = 0.$$

0 is the vector having all zero entries, i.e., $0 = [0, \dots, 0]^T$

■ B is a basis for S if:

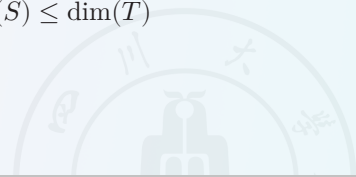
1 $L(B) = S$, i.e., B spans S

2 B is linearly independent



A Review of Linear Algebra: Subspace

- Theorem 1: Every subspace, S of $\mathbb{R}^{n \times 1}$, has a basis. Any two bases of a subspace must contain the same number of elements (Dimension)
- Theorem 2: Every set of vectors that spans a subspace contains a basis for the subspace
- Theorem 3: Any linearly independent subset of a subspace can be extended to a basis for the space
- Theorem 4: If S is a subspace of $\mathbb{R}^{n \times 1}$ with $\dim(S) = k$, then any subset of S that contains more than k vectors is a linearly dependent set of vectors
 - If S and T are both subspace of $\mathbb{R}^{n \times 1}$, and $S \subseteq T$, then $\dim(S) \leq \dim(T)$



A Review of Linear Algebra: Subspace

- An $m \times n$ real matrix A :

$$A = [a_{i,j}] = \begin{bmatrix} a_{1,1} & \cdots & a_{1,j} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots & & \vdots \\ a_{i,1} & \cdots & a_{i,j} & \cdots & a_{i,n} \\ \vdots & & \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,j} & \cdots & a_{m,n} \end{bmatrix}$$

- The j -th column of A : $A^{(j)} = [a_{1,j}, \dots, a_{i,j}, \dots, a_{m,j}]^T$
- The i -th row of A : $A_{(i)} = [a_{i,1}, \dots, a_{i,j}, \dots, a_{i,n}]$



A Review of Linear Algebra: Subspace

- Column space: The space spanned by the columns of A , which is a subspace of $\mathbb{R}^{m \times 1}$
- Row space: The space spanned by the rows of A , which is a subspace of $\mathbb{R}^{n \times 1}$
- Rank: The dimension of the column (row) space of A is called the column (row) rank of A :

$$\text{rank}(A) = \text{rank}_{\text{col}}(A) = \text{rank}_{\text{row}}(A)$$



A Review of Linear Algebra: Matrix Operation

- Matrix addition:

$$A + B = [a_{i,j} + b_{i,j}]$$

- Matrix scaling:

$$\alpha A = [\alpha a_{i,j}]$$

- Matrix product: $A = [a_{i,j}] \in \mathbb{R}^{m \times k}$, $B = [b_{i,j}] \in \mathbb{R}^{k \times n}$

$$AB = \left[\sum_{l=1}^k a_{il} b_{lj} \right] \in \mathbb{R}^{m \times n}$$



A Review of Linear Algebra: Identity Matrix

■ Identity Matrix:

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} = [\delta_{i,j}] \in \mathbb{R}^{n \times n}$$

■ Kronecker delta:

$$\delta_{i,j} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$



A Review of Linear Algebra: Submatrix

■ B is a submatrix of A :

$$A = \begin{bmatrix} 6 & 1 & 3 & 4 & 0 & 5 \\ 2 & 1 & 4 & 0 & 0 & 1 \\ 1 & 0 & 1 & -1 & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}.$$



A Review of Linear Algebra: Blocks

- Blocks: The submatrices in a partitioned matrix

$$A = \begin{bmatrix} 6 & 1 & 3 & 4 & 0 & 5 \\ 2 & 1 & 4 & 0 & 0 & 1 \\ 1 & 0 & 1 & -1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} B & D & O & b \\ -c_B^T & -c_D^T & 1 & 0 \end{bmatrix},$$

$$B = \begin{bmatrix} 6 & 1 \\ 2 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 3 & 4 \\ 4 & 0 \end{bmatrix}, \quad O = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad b = \begin{bmatrix} 5 \\ 1 \end{bmatrix},$$

$$-c_B^T = [1 \ 0], \quad -c_D^T = [1 \ -1].$$

A Review of Linear Algebra: Blocks

- Matrix multiplication using blocks as elements

$$A = \begin{bmatrix} C & D \\ E & F \end{bmatrix}, \quad B = \begin{bmatrix} G & H & J \\ K & L & M \end{bmatrix}$$

$$AB = \begin{bmatrix} CG + DK & CH + DL & CJ + DM \\ EG + FK & EH + FL & EJ + FM \end{bmatrix}.$$



A Review of Linear Algebra: Matrix Inverse

- Two square matrices:

$$A = [a_{i,j}] \in \mathbb{R}^{n \times n}, \quad B = [b_{i,j}] \in \mathbb{R}^{n \times n}$$

- $AB = BA = I_n$:
 A is nonsingular, $B = A^{-1}$ is the inverse of A
- A NOT invertible is singular.



A Review of Linear Algebra: Matrix Inverse

- $A = [a_{i,j}] \in \mathbb{R}^{n \times n}$, A is nonsingular:

$$\text{rank}(A) = n \iff \text{columns (rows) are linearly independent}$$

- Columns are linearly dependent, so singular:

$$\begin{bmatrix} 1 & 1 & 4 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

- Columns are linearly independent, so nonsingular:

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$



A Review of Linear Algebra: Elementary Row Operations

- Type 1: $\alpha A_{(i)}$, $\alpha \neq 0$
- Type 2: $A_{(i)} + \alpha A_{(j)} \rightarrow A_{(i)}$
- Type 3: $A_{(i)} \rightleftarrows A_{(j)}$
- Elementary row operations do not change the row space of a matrix

$$[A \mid I_n] \longrightarrow \cdots \longrightarrow [I_n \mid A^{-1}]$$

- If the above is impossible, then A is singular



A Review of Linear Algebra: Elementary Row Operations

■ Nonsingular $B \in \mathbb{R}^{m \times m}$, $-c_B^\top \in \mathbb{R}^{1 \times m}$, $0 = [0, \dots, 0]^\top$

$$\begin{bmatrix} B^{-1} & 0 \\ c_B^\top B^{-1} & 1 \end{bmatrix} \begin{bmatrix} B & 0 \\ -c_B^\top & 1 \end{bmatrix} = \begin{bmatrix} I_m & 0 \\ 0^\top & 1 \end{bmatrix} = I_{m+1}.$$

$$\Rightarrow \begin{bmatrix} B & 0 \\ -c_B^\top & 1 \end{bmatrix}^{-1} = \begin{bmatrix} B^{-1} & 0 \\ c_B^\top B^{-1} & 1 \end{bmatrix}, \quad \text{i.e., nonsingular.}$$



A Review of Linear Algebra: Elementary Row Operations

- With a specific sequence of elementary row operations

$$[B \mid I_m] \rightarrow [I_m \mid B^{-1}]$$

- With the same sequence of elementary row operations

$$\left[\begin{array}{cc|cc} B & 0 & I_m & 0 \\ -c_B^T & 1 & 0^T & 1 \end{array} \right] \rightarrow \left[\begin{array}{cc|cc} I_m & 0 & B^{-1} & 0 \\ -c_B^T & 1 & 0^T & 1 \end{array} \right]$$

- Replace by “the last row” $+c_B^T \cdot$ “the block row”

$$\left[\begin{array}{cc|cc} I_m & 0 & B^{-1} & 0 \\ -c_B^T & 1 & 0^T & 1 \end{array} \right] \rightarrow \left[\begin{array}{cc|cc} I_m & 0 & B^{-1} & 0 \\ 0^T & 1 & c_B^T B^{-1} & 1 \end{array} \right].$$



Example 8

■ Question: Solve $Ax = b$ for x where

$$A = \begin{bmatrix} 1 & 1 & -1 \\ -2 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}.$$

Example 8

■ Solution:

$$\left[\begin{array}{ccc|ccc|c} 1 & 1 & -1 & 1 & 0 & 0 & 1 \\ -2 & 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \end{array} \right]$$

$$\xrightarrow{A_{(2)}+2A_{(1)}, A_{(3)}-A_{(1)}} \left[\begin{array}{ccc|ccc|c} 1 & 1 & -1 & 1 & 0 & 0 & 1 \\ 0 & 3 & -1 & 2 & 1 & 0 & 4 \\ 0 & 0 & 2 & -1 & 0 & 1 & -1 \end{array} \right]$$

$$\xrightarrow{A_{(1)}-\frac{1}{3}A_{(2)}, \frac{1}{3}A_{(2)}} \left[\begin{array}{ccc|ccc|c} 1 & 0 & -\frac{2}{3} & \frac{1}{3} & -\frac{1}{3} & 0 & -\frac{1}{3} \\ 0 & 1 & -\frac{1}{3} & \frac{2}{3} & \frac{1}{3} & 0 & \frac{4}{3} \\ 0 & 0 & 2 & -1 & 0 & 1 & -1 \end{array} \right]$$

$$\xrightarrow{A_{(1)}+\frac{1}{3}A_{(3)}, A_{(2)}+\frac{1}{6}A_{(3)}, \frac{1}{2}A_{(3)}} \left[\begin{array}{ccc|ccc|c} 1 & 0 & 0 & 0 & -\frac{1}{3} & \frac{1}{3} & -\frac{2}{3} \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{3} & \frac{1}{6} & \frac{7}{6} \\ 0 & 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} & -\frac{1}{2} \end{array} \right].$$

$$A^{-1} = \begin{bmatrix} 0 & -\frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}, \quad x = \begin{bmatrix} -\frac{2}{3} \\ \frac{7}{6} \\ -\frac{1}{2} \end{bmatrix}.$$

A Review of Linear Algebra: Elementary Row Operations

- Augmented matrix: $[A \mid b]$
- Gauss-Jordan method of elimination: Solving $Ax = b$ by elementary row operations, to obtain $[I_m \mid A^{-1}b]$
- Elementary matrix: Performing the elementary row operations on an identity matrix

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}.$$

A Review of Linear Algebra: Gaussian Elimination

- Gaussian elimination: Applying forward elimination and back substitution together to solve $Ax = b$
 - Upper triangular matrix: U , all zeros below the main diagonal
 - Forward elimination: Using elementary row operations on $[A \mid b]$ to obtain $[U \mid \bar{b}]$
 - Back substitution: Substituting $x_m, x_{m-1}, \dots, x_{m-i}$ into the $(m-i-1)$ -th equation and solving for x_{m-i-1}
- Question: Solve linear system with Gaussian elimination

$$2x_2 - x_3 = 8,$$

$$x_1 + 2x_2 = 6,$$

$$3x_2 + 4x_3 = 12.$$



A Review of Linear Algebra: Gaussian Elimination

- Solution: forward elimination on the augmented matrix $[A \mid b]$

$$\begin{bmatrix} 0 & 2 & -1 & 8 \\ 1 & 2 & 0 & 6 \\ 0 & 3 & 4 & 12 \end{bmatrix} \xrightarrow{A_{(1)} \leftrightarrow A_{(2)}} \begin{bmatrix} 1 & 2 & 0 & 6 \\ 0 & 2 & -1 & 8 \\ 0 & 3 & 4 & 12 \end{bmatrix}$$

$$\xrightarrow{A_{(2)} \leftrightarrow A_{(3)}} \begin{bmatrix} 1 & 2 & 0 & 6 \\ 0 & 3 & 4 & 12 \\ 0 & 2 & -1 & 8 \end{bmatrix} \xrightarrow{A_{(3)} - \frac{2}{3}A_{(2)}} \begin{bmatrix} 1 & 2 & 0 & 6 \\ 0 & 3 & 4 & 12 \\ 0 & 0 & -\frac{11}{3} & 0 \end{bmatrix}.$$

By back substitution, $x_3 = 0$, $x_2 = 4$, and $x_1 = -2$.



A Review of Linear Algebra: Gaussian Elimination

■ The upper triangular matrix

$$U = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & -\frac{11}{3} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{2}{3} & 1 \end{bmatrix}}_{F_3} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}}_{F_2} \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{F_1} A.$$

$$F_2 F_1 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$



A Review of Linear Algebra: LU-decomposition

- Row echelon form: Upper triangular (for square matrix)
 - Pivot: The leftmost nonzero entry of a row
 - Pivoting: Elementary row operations to find pivot
 - Pivot matrix: Matrix that performs a row interchange; it is a permutation matrix obtained by applying the same interchange to I_m
- Permutation matrix: the rows of I_m are permuted



A Review of Linear Algebra: LU-decomposition

- Apply Gaussian elimination to $Ax = b$ for row echelon form:

$$Ux = (F_k P_k) \cdots (F_1 P_1) Ax = (F_k P_k) \cdots (F_1 P_1) b = \bar{b}.$$

- Collecting the row permutations into a permutation matrix P and the elimination operations into a lower-triangular matrix L gives

$$PA = LU.$$

- Hence

$$Pb = PAx = LUx = Ly,$$

followed by $Ux = y$.



Affine & Convex Sets

■ For $x_i \in \mathbb{R}^{n \times 1}$, $\forall i = 1, \dots, k$

■ Linear combination:

$$\sum_{i=1}^k \alpha_i x_i, \quad \forall \alpha_i \in \mathbb{R}$$

■ Affine combination:

$$\sum_{i=1}^k \alpha_i x_i, \quad \sum_{i=1}^k \alpha_i = 1$$

■ Conical (nonnegative) combination:

$$\sum_{i=1}^k \alpha_i x_i, \quad \forall \alpha_i \geq 0$$

■ Convex combination:

$$\sum_{i=1}^k \alpha_i x_i, \quad \forall \alpha_i \geq 0 \cap \sum_{i=1}^k \alpha_i = 1$$

Affine & Convex Sets

- $S \subseteq \mathbb{R}^{n \times 1}, S \neq \emptyset$

- Linear hull of S :

$$L(S) = \left\{ \sum_{i=1}^k \alpha_i x_i \mid \alpha_i \in \mathbb{R}, x_i \in S, k \in \mathbb{N} \right\}$$

- Affine hull of S :

$$A(S) = \left\{ \sum_{i=1}^k \alpha_i x_i \mid \alpha_i \in \mathbb{R}, \sum_{i=1}^k \alpha_i = 1, x_i \in S, k \in \mathbb{N} \right\}$$

Affine & Convex Sets

- Conical hull of S :

$$C(S) = \left\{ \sum_{i=1}^k \alpha_i x_i \mid \alpha_i \geq 0, x_i \in S, k \in \mathbb{N} \right\}$$

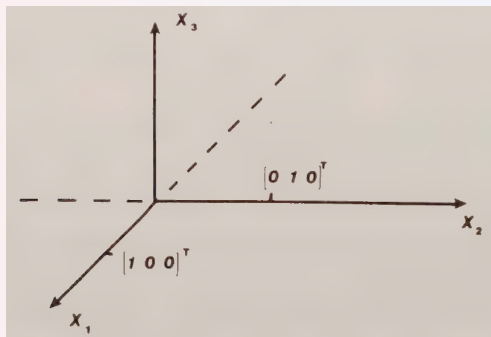
- Convex hull of S :

$$X(S) = \left\{ \sum_{i=1}^k \alpha_i x_i \mid \alpha_i \geq 0, \sum_{i=1}^k \alpha_i = 1, x_i \in S, k \in \mathbb{N} \right\}$$



Example 9

- Question: $S = \{[1, 0, 0]^T, [0, 1, 0]^T\}$, determine $L(S)$, $A(S)$, $C(S)$, and $X(S)$
- Solution:



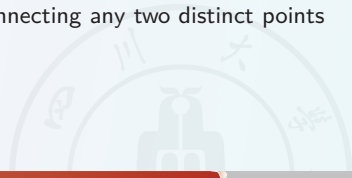
Affine & Convex Sets

- Proposition 1: For any nonempty set $S \subseteq \mathbb{R}^{n \times 1}$,

$$S \subseteq X(S) \subseteq A(S) \subseteq L(S),$$

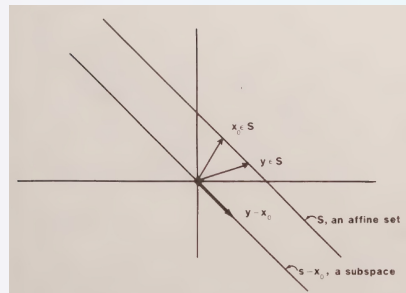
$$S \subseteq X(S) \subseteq C(S) \subseteq L(S).$$

- Proposition 2: S is a subspace of $\mathbb{R}^{n \times 1} \iff S = L(S)$
- Proposition 3: S is an affine set $\iff S = A(S) \iff$ the line passing through any two distinct points in S is a subset of S
- Proposition 4: S is a convex cone $\iff S = C(S) \iff$ every nonnegative linear combination of points in S belongs to S
- Proposition 5: S is a convex set $\iff S = X(S) \iff$ the line segment connecting any two distinct points in S is a subset of S



Affine & Convex Sets

- Proposition 6: A subset $S \subseteq \mathbb{R}^{n \times 1}$ is affine $\iff S - x_0$ is a subspace of $\mathbb{R}^{n \times 1}$, for every $x_0 \in S$
- An affine set is just the translate of a subspace
- $\dim(S) = \dim(S - x_0)$
- For any subset $T \subseteq \mathbb{R}^{n \times 1}$,
 $\dim(T) = \dim(A(T))$



Affine & Convex Sets

For $c \neq 0$ and $\beta \in \mathbb{R}$:

■ Hyperplane:

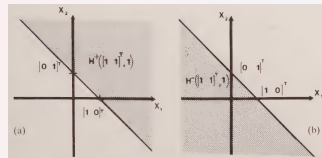
$$H(c, \beta) = \{x : c^T x = \beta\}$$

■ Upper half space:

$$H^+(c, \beta) = \{x : c^T x \geq \beta\}$$

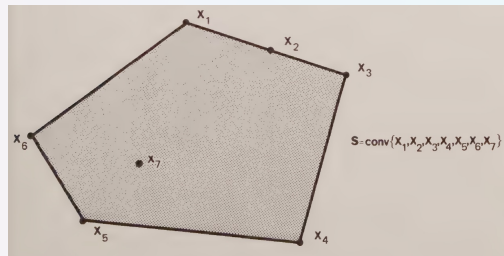
■ Lower half space:

$$H^-(c, \beta) = \{x : c^T x \leq \beta\}$$



Affine & Convex Sets

- Convex Polytope: $X(\{x_1, x_2, \dots, x_k\})$, $k < +\infty$
- Affinely Independent: $\{x_0, x_1, \dots, x_k\}$ is affinely independent $\iff \{x_1 - x_0, x_2 - x_0, \dots, x_k - x_0\}$ is linearly independent
- k -dimensional simplex: A convex polytope which is a convex hull of $k + 1$ affinely independent vectors



Affine & Convex Sets

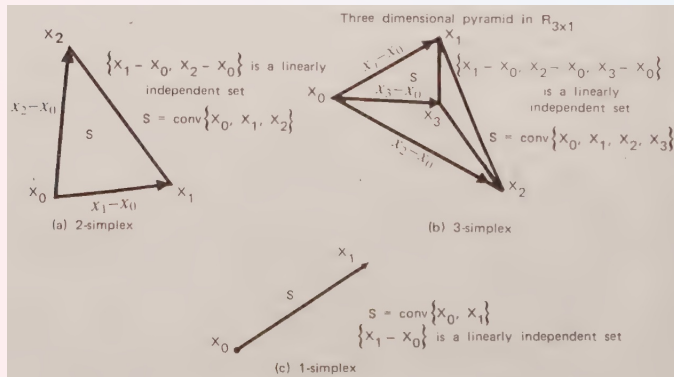
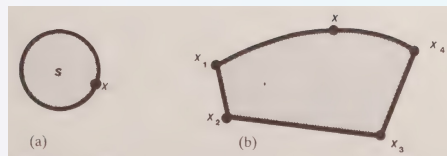


Figure 1: (a) 2-simplex, (b) 3-simplex, (c) 1-simplex

Affine & Convex Sets

- Extreme Point: $z \in S$ (convex set) is an extreme point of $S \iff$ there does NOT exist distinct $x, y \in S$ and $0 < \alpha < 1$ such that $z = \alpha x + (1 - \alpha)y$
- The vertices of any simplex are extreme points of the simplex
- The extreme points of a convex set S are those NOT on the interior of any line segment connecting any other pair of points of S



Linear Programming Problem

- Original expression

$$\min c_1x_1 + \dots + c_nx_n$$

subject to

$$a_{1,1}x_1 + \dots + a_{1,n}x_n \leq b_1,$$

$$\vdots$$

$$a_{m,1}x_1 + \dots + a_{m,n}x_n \leq b_m,$$

and

$$\forall x_i \geq 0.$$



Linear Programming Problem

- Equivalent expression:

$$\min c^T x$$

subject to

$$\begin{aligned} Ax &= A^{(1)}x_1 + A^{(2)}x_2 + \dots + A^{(n)}x_n \leq b, \\ x &\geq 0. \end{aligned}$$

- Set of feasible solutions: $F = \{x : Ax \leq b \cap x \geq 0\}$

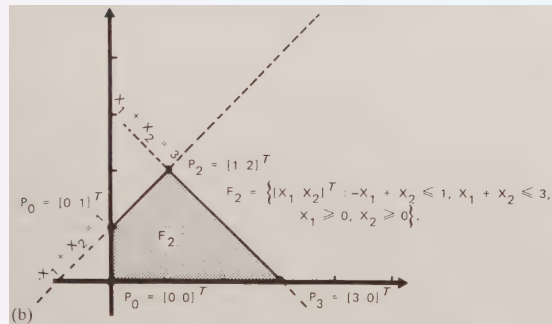
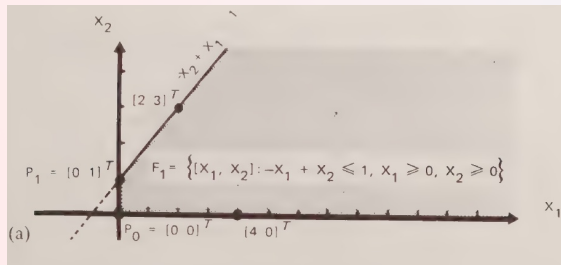


Linear Programming Problem

- Proposition 7: The set of feasible solution, F , to the linear programming problem is a convex set
- Proof: Let $x_1, x_2 \in F$, $\alpha \in [0, 1]$, check the convex combination $z = \alpha x_1 + (1 - \alpha)x_2$. Does $z \in F$?
- $F = \left(\bigcap_{i=1}^m \{x : A_{(i)}x \leq b_i\} \right) \cap \left(\bigcap_{i=1}^n \{x : x_i \geq 0\} \right)$
- Convex polyhedron: Intersection of finite number of half spaces
- Convex polytope $\implies$ Convex polyhedron; the converse does not hold in general



Linear Programming Problem



Linear Programming Problem

Finite Basis Theorem

Let $F = \{x \in \mathbb{R}^{n \times 1} : Ax \leq b\}$ be a nonempty convex polyhedron. Then there exist finite sets $\{x_1, \dots, x_d\}$ and $\{y_1, \dots, y_\ell\}$ such that

$$F = X(\{x_1, \dots, x_d\}) + C(\{y_1, \dots, y_\ell\}),$$

i.e.,

$$F = \left\{ \sum_{i=1}^d \alpha_i x_i + \sum_{j=1}^{\ell} \beta_j y_j : \alpha_i \geq 0, \beta_j \geq 0, \sum_{i=1}^d \alpha_i = 1 \right\}.$$



Linear Programming Problem

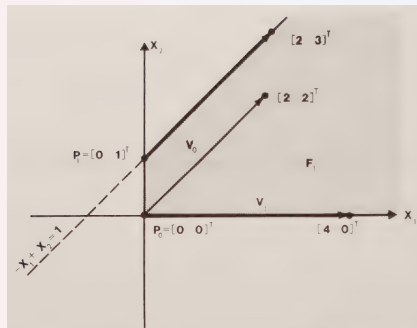


Figure 3: $F_2 = X(\{p_0, p_1\}) + C(\{v_0, v_1\})$, i.e., F_2 can be expressed as $x = \alpha_0 p_0 + \alpha_1 p_1 + \beta_0 v_0 + \beta_1 v_1$, where $\alpha_0 \geq 0$, $\alpha_1 \geq 0$, $\beta_0 \geq 0$, $\beta_1 \geq 0$ and $\alpha_0 + \alpha_1 = 1$